A64 imported from www.frozencpu.com. The thing cost him $60 (Rs 2,700), including the shipping charges. With a water block in place, Sunny was all geared to rig up his own cooling system.

What's In A Block?

A water block, the water cooling equivalent of a heat sink, is used on several computer components including the CPU, graphics card, and the chipset on the motherboard. A typical water block consists of two parts. The first is the base, which makes contact with the device being cooled. This is usually made of a highly-conductive metal, such as aluminium or copper, so that heat is dissipated fast. The second part—the top—ensures that the water (or the liquid used for cooling) is contained safely inside the block.

If you want to jazz up your PC, you can use a loop made of a transparent material such as Perspex, or UV-reactive tubes that glow under UV (ultraviolet) light.

The base and top are sealed together to form a block with a channel inside. Each end of the channel has connectors for the inlet and outlet. In most cases, the channel is a spiral or zig-zag in pattern, like in a refrigerator or air-conditioner.

Some channels also have heat sink-styled fins so as to enlarge the surface area available for heat dissipation. Care must be taken, however, so the design of the channel does not impede the flow of the liquid inside it. Most people who opt for water cooling prefer to have the channels made of a transparent material so that the flow of the liquid inside is visible. Guess it adds to the rush of things!

The Geek Setup

Sanjay's cooling system consisted of the water block and a pump that pumped liquid at a speed of 700 litres per hour. A stroke of genius was in picking up a radiator from a Mitsubishi Lancer. "The radiator was one foot high and 20 cm wide. The dimensions were just right for my design," claims Sanjay. "I also needed a few tubes to create the channels, some coolant, and water. My hunt for the channels was the most exciting part. I was looking for a UV-reactive tube, and these weren't available in India. A little bit of reading up made me realise that these tubes were indeed available here, only they were used for different purposes.

"These tubes were used in motorbikes. Later on, people were amazed as to how I laid my hands on them! The other important components were the coolant and the distilled water. For the coolant, I used a regular engine coolant from

Check out the rig—with the water block, neon lighting and all

Castrol. I did not, however, trust the purity of the distilled water found in garages, so I borrowed two litres of it from my college laboratory. I, then mixed one part of coolant with four parts of distilled water. I connected the rig, screwed in the right parts, and voila—my water cooling setup was ready," he explains.

To test his cooling system, Sanjay did some volt modding—increasing the voltage intake of his motherboard—on his A7N8X Dlx nForce2 Ultra motherboard. The cooling setup enabled him touch a core voltage of 2.05 volts. The resulting heat dissipation was around 146 watts. Compare this to the normal heat dissipation of 65 watts at a core voltage of 1.65 volts, and you know how much cooling the system was doing! Of course, running a motherboard at that voltage for too long meant that you would need to replace it within a few hours, so Sanjay settled down at a core voltage of 1.7 volts, which is still significantly higher than the normal 1.65 volts.

Let's Talk About The Money

Sanjay's constant experiments have led to motherboard and processor burnouts. Not to say, they've also burnt a hole in his pocket, but this hasn't deterred this Information Science student from experimenting further.

His aim now is to design and build his own water block. "Once my exams are done, I'll get down to developing my own water block and make a Do-It-Yourself kit. I plan to sell these for Rs 4,000. That is significantly lower than what most water cooling systems cost today," he says.

Not for the faint-hearted or even the moderately geeky, water cooling will remain niche because of the level of customisation it requires. However, you do get computers that have a liquid cooling system installed, notably the Apple G5. Remember, you can't just slap a water block on your processor and overclock it! If you're interested in water cooling, make sure you know what you're doing! ⊠

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Sanjay Rao, the whiz who did it